

Modular Smart Vending Cabinet with Smart-Glass Inspection Mode, Sensor-Fusion Manual Retrieval, and Quality-Based Dynamic Pricing

Field of the Invention

The present invention relates to autonomous retail cabinets and vending systems, and more particularly to a modular smart vending cabinet having a unified chassis supporting interchangeable dispensing and sensing modules, including privacy-first smart-glass inspection access for restricted goods, sensor-fusion based pay-as-you-pick charging for standard goods, and AI-driven revenue management for perishable and quality-sensitive products.

Background

Conventional vending machines and automated retail cabinets typically fall into one of several categories: (i) age-gated or ID-scanning dispensers that block dispensing until a credential is scanned; (ii) robotic vending cabinets that use a gantry or robotic retrieval mechanism; and (iii) dynamic pricing systems that apply discounts based on simple rules such as printed expiration dates or environmental conditions. These systems often do not provide a privacy-first, compliant user inspection experience for restricted goods, do not support a unified chassis that enables direct manual retrieval across multiple module types without robotics, and do not implement revenue-management algorithms that discount based on modeled sensory quality degradation and on slot-capacity yield (space yield) rather than only time-to-expiration. These systems also generally lack a transformer-style chassis with a vertical power busbar for rapid module reconfiguration, integrated mobility for rear-loading FIFO service, sales-final anti-reverse enforcement, automated purge-to-waste handling, and networked mobile experiences that expose item-level expiration and purge timing.

Accordingly, there is a need for an integrated, hardware-anchored modular cabinet that (a) supports multiple dispensing modalities within a unified chassis; (b) enables direct manual retrieval monitored by a sensor-fusion network (rather than robotic retrieval); (c) implements a privacy-first restricted-goods workflow using smart glass and a digital information display prior to access; and (d) applies dynamic pricing based on sensory decay modeling and space-yield displacement to reduce waste and increase revenue.

Brief Description of Drawings

FIG. 1 is a front view of an autonomous retail cabinet showing the overall hardware layout, including the door, user-facing display, biometric/age-verification camera, payment interface, and the internal smart shelves. The figure emphasizes the physical positioning of the camera and the linear “front zone” antenna region along the shelf edge that together support sensor-fusion-based anti-theft monitoring.

FIG. 2 is a side or sectional view of the cabinet illustrating the internal electronics bay and offline architecture. The figure depicts the edge controller, local database or cache, communication modules, and backup power supply, showing how core functions (door control, counting items, and local transaction buffering) are implemented so that the system can continue to operate even when an external network or cloud AI service is unavailable.

FIG. 3 is a detail view of the shelf and sensing geometry showing the linear front-zone antenna array and the camera field of view. The figure highlights the overlap region between the camera's viewing cone and the antenna detection zone used for sensor fusion, including the configuration in which a hand entering the field of view without a corresponding RFID removal event triggers an alarm or deposit lock.

FIG. 4 is a high-level process diagram illustrating the induction and stocking workflow and its relationship to the hardware configuration. The figure shows an induction station or handheld scanner used by staff to scan a product barcode, scan or program the RFID tag, and input the expiration date, thereby associating SKU and expiration data with a unique RFID identifier prior to stocking items on the shelves monitored by the front-zone antenna array and camera.

Detailed Description

In one embodiment, and with reference to FIG. 1, a **universal smart cabinet** system comprises a secure enclosure having an interior volume accessible through at least one door, an access-control mechanism coupled to the door, a **unified reconfigurable chassis** arranged within the interior volume, an exterior user interface, and one or more cameras mounted on or near the front of the cabinet. The universal smart cabinet may be implemented as one or more sealed unit types including **(i) a secure unit** for restricted goods, **(ii) an ambient unit**, and **(iii) a chilled unit** (e.g., refrigerated), each sharing the same chassis architecture and sensor-fusion network but operating under different access-control and transaction state machines. The exterior user interface may include a touch-sensitive display and/or physical buttons that allow a user to authenticate, request cabinet access, view pricing information, and confirm purchase operations. In secure mode embodiments, the cabinet further includes a biometric camera (also referred to as "The Eye") positioned to capture facial image data of users in front of the cabinet and, in some implementations, also observe an interior region near the front of the chassis modules. In embodiments, the front edge of one or more modules defines a linear "front zone" region where a corresponding antenna matrix is positioned, as generally illustrated in FIG. 1.

I. Hardware Chassis (Transformer Base). In embodiments, the unified chassis includes a vertical power busbar disposed along a rear wall of the enclosure, the busbar providing conductive rails and mating connectors configured to deliver power and data to shelf modules when docked at selectable vertical positions, thereby enabling rapid vertical reconfiguration of modules without rewiring. In some implementations, module control signals are exchanged via a wired bus through the busbar and/or a short-range wireless link between the edge controller and the modules.

In embodiments, the cabinet further includes a mobility system comprising industrial casters configured to support rolling movement of the cabinet and retractable leveling feet configured to

engage a floor surface to stabilize the cabinet during use. The mobility system enables an operator to undock the cabinet from a wall or fixed position to access a rear-loading side of the modules for FIFO stocking and service.

In embodiments, each module includes a smart edge along a user-facing front portion comprising a high-density antenna matrix (rather than a single antenna) and an LED information strip configured to provide navigation and pricing cues, including per-item indicators, guidance to discounted items, and user prompts during a session.

With reference to FIG. 2, the system further includes a processing system implemented as an edge controller or edge AI unit located in an electronics bay within or adjacent to the cabinet. The edge controller includes at least one processor, memory, and one or more machine-readable instruction sets that implement a pricing engine, an inventory-valuation module, a biometric analysis module, an access-control module, a transaction-processing module, and a sensor-fusion module. The edge controller is communicatively coupled to the various sensors (including the cameras, RFID antenna matrix, and, where installed, load cells), to the access-control mechanism, and, when available, to a remote server via a network interface (e.g., LTE, Wi-Fi, or Ethernet). As shown schematically in FIG. 2, the edge controller maintains a local cache or database of inventory, pricing, user deposit, and transaction data, and may be powered in part by a backup or uninterruptible power supply so that core functions can continue during power or network disruptions. In particular, basic functions such as unlocking and locking the door, executing the cabinet state machine, detecting removals, updating inventory, and debiting funds against a locally maintained deposit or balance can be carried out autonomously via local caching on the edge controller even when a cloud AI service or external network connection is unavailable, thereby preventing the cabinet from becoming inoperative in the event of an internet outage.

With reference to FIG. 3, the unified chassis is configured to simultaneously support multiple interchangeable mechanical modules for direct manual retrieval, including at least: (i) smart pushers for bottles/cartons, (ii) smart hooks or rods for hanging items, (iii) box-on-scale zones for soft or fragile items, (iv) gated gravity-feed channels for small or low-cost items, (v) sliding tray modules comprising low-friction gravity shelves without springs for soft packaging (e.g., meat, cheese, deli), and (vi) auto-purge modules configured to expel expired or quarantined items to a smart waste compartment. In an example configuration, a pusher module comprises spring-loaded tracks and movable dividers (adjustable width) configured to advance products toward a front retrieval edge, and the front edge includes an antenna matrix configured for virtual zoning and tag tracking. A hook module comprises hanging rods or hooks that are configured for rear loading and direct manual removal from a front access area. A box-on-scale module comprises a designated box or zone supported by at least one load cell, where removal is confirmed by a weight delta and corroborated by camera evidence, thereby accommodating items not well-suited to compression by springs (e.g., milk pouches, mayonnaise jars, coffee beans in boxes) and enabling multi-picking during a session. A gated gravity-feed module comprises an inclined shelf or chute configured to deliver items to a bottom weighing dish supported by a load cell and includes a controllable gate for sequential release.

In embodiments, the system enforces **FIFO loading and rotation** as part of an operational workflow. With reference to FIG. 4, the system includes an induction station used for associating product information with internal tracking tags prior to stocking the cabinet. The induction station may be embodied as a fixed docking station or as a handheld scanner device that is communicatively coupled to the edge controller. In an example induction workflow, staff scan a manufacturer barcode or SKU code to register an item, and then apply an internal tracking tag (e.g., RFID, NFC, or a machine-readable sticker) that is associated in a database with the SKU and, where applicable, a lot/batch identifier and an expiration date or production date. The cabinet is configured for **rear loading** such that new stock is inserted from the back of a module and pushes existing stock toward the front retrieval edge, thereby ensuring that the first item loaded is the first item sold (first-in-first-out). This pre-association and rear-loading configuration ensures that, when a tagged item is later detected by the antenna matrix and/or other sensors, the edge controller can resolve product identity and related metadata from the tag association.

By combining the pre-association performed at the induction station (FIG. 4) with per-module sensing and camera overlap (FIG. 3), the system maintains an accurate item-level and/or position-level view of inventory and transactions. Unlike robotic vending cabinets that rely on gantries or robotic retrieval mechanisms, the claimed system supports **direct manual retrieval** and uses a unified **sensor-fusion region** to confirm removal events across module types. In embodiments, the sensor-fusion module combines at least two of: (i) RFID readings from the antenna matrix or segmented near-field antennas, (ii) camera detections of hand entry, product motion, and removal gestures, (iii) weight delta measurements from load cells in box-on-scale zones and/or weighing dishes, and (iv) door state and time-window logic. For example, if the camera detects that a user's hand has removed an object from a particular zone within the overlap region but the corresponding RFID antennas do not detect removal of an expected tag (e.g., due to RF shielding by foil packaging), the system treats this as an anomalous event. In response, the edge controller may trigger an audible or visual alarm, temporarily restrict access, hold or increase a user deposit, flag the session for review, or otherwise apply anti-fraud controls.

In embodiments employing the gated gravity-feed module, the system implements a **gated gravity sequencing logic** in which release of a next item is dependent on confirmed removal of a prior item. In one implementation, a first item sits in a weighing dish at the bottom of the chute and the gate remains locked while the load cell detects the presence of the item (weight above a threshold). When the user removes the first item, the system detects the weight returning to zero (or below a removal threshold) and corroborates removal via camera identification of the item and/or the removal gesture. **Only after** the removal is confirmed does the edge controller actuate the gate to release a second item into the weighing dish, after which the gate is again locked until the second item is removed. This gating logic prevents sweeping multiple items without detection and enables accurate per-item charging in low-cost or high-velocity item lanes.

In embodiments employing a sliding tray module, the chassis supports one or more gravity shelves having a low-friction floor and a user-facing opening, wherein products are advanced toward the opening by gravity and/or by user movement rather than by spring compression. The

sliding tray module is configured for soft or deformable packaging that would be damaged by spring-loaded pushers, including meat, cheese, deli items, and other flexible packs.

IV. “No Return” Policy (Anti-Reverse System). In embodiments, the system enforces a sales-final policy using at least one of (i) mechanical anti-reverse constraints and (ii) software-based penalties. In a mechanical embodiment, pusher modules and hook modules include a one-way ratchet or stop feature configured to prevent rearward movement of products once advanced toward the retrieval edge. In a software embodiment used for weight-based zones (including scale-box zones, bins, and trays), the controller monitors load-cell readings and, upon detecting a weight increase indicative of a return event during an active session, applies a configured penalty charge, including a double-charge penalty relative to a current unit price. In such embodiments, the user interface presents a mandatory consent control (e.g., an “I AGREE” button) acknowledging the penalty policy prior to unlocking the door.

V. Auto-Purge & Smart Waste. In embodiments, the cabinet automatically clears expired or quarantined items to free slot capacity. In one implementation, a module includes a retractable front stop that is movable between a retain position and a purge position. When a purge condition is met (e.g., scheduled purge time, operator command, or expiration status), the controller actuates the retractable stop to release the retained item, and a spring or gravity force advances the item forward such that the item drops into a bottom waste compartment. The waste compartment is implemented as a smart waste bin equipped with at least one load cell and an RFID antenna, and items deposited therein are tracked as quarantined rather than deleted, pending staff collection. To discourage scavenging, the controller may apply an anti-scavenger rule in which retrieval of an item from the smart waste bin triggers a charge at a full original price associated with the item.

VI. Interface & Transparency. In embodiments, prior to a payment commitment or door unlock, the cabinet display and/or a companion mobile application presents item-specific information for physical units inside the cabinet, including an exact expiration date and a scheduled purge time for at least a subset of items. In embodiments, the transaction module applies a real-time incremental hold in which funds are authorized or blocked on a payment instrument per item removed, substantially in real time, rather than aggregating a single sum at the end of the session.

In one embodiment, the universal smart cabinet operates under multiple logic flows depending on unit type. In an embodiment directed to restricted goods such as alcohol (Unit A: Secure), the system operates in a restricted-mode configuration that includes an opaque smart glass door panel and a “digital sommelier” user experience. In such embodiments, the access-control mechanism remains locked until completion of a biometric age-verification protocol and a payment commitment (e.g., a pre-authorization hold). In embodiments, the restricted unit includes a smart glass (e.g., PDLC) panel configurable for at least an opaque sleep mode and a transparent inspection mode, and the restricted unit employs a mandatory triple sensor-fusion configuration combining (i) the antenna matrix, (ii) an overhead or interior-facing camera, and (iii) load cells associated with at least one restricted-goods zone. The load cells enable detection of partial-consumption and return events (e.g., opening or drinking from a container and returning it) and reduce fraud scenarios such as tag swapping or “ghost” returns. The smart glass defaults to an opaque state to enforce privacy and prevent visual access to restricted goods.

When a user approaches the cabinet, the biometric camera captures facial image data and the biometric analysis module executed on the edge controller runs one or more models to estimate age and/or match the face to a verified identity associated with a user account or backend KYC database. Upon successful verification (e.g., age greater than or equal to a legal minimum age such as twenty-one years, optionally with confidence above a threshold), the system enters an inspection mode in which the smart glass transitions from opaque to transparent and the display presents product information (e.g., vintage, tasting notes, origin, or pairing suggestions) prior to granting physical access. Following inspection mode, the transaction-processing module initiates a payment commitment logic flow, such as placing a pre-authorization hold or deposit, optionally based on a risk profile. Only upon confirmation of the payment commitment does the access-control module release the door to permit manual retrieval. Upon removal, the sensor-fusion module confirms the removed item(s), and the system captures the corresponding charge, releasing any remaining pre-authorization when applicable.

In an embodiment directed to ambient and chilled sealed units (Unit B: Standard), the system uses a transparent door panel and a tap-to-pay or equivalent authorization flow. Upon receiving a payment authorization, the access-control mechanism unlocks and the system operates in a **pay-as-you-pick** configuration in which each item removal is detected by sensor fusion (e.g., RFID plus camera, and load-cell deltas where present) and charged immediately or near-immediately. The user interface may provide a digital label function in which product information and pricing (including discounts) are displayed and can be zoomed or browsed during the session.

In embodiments, the pricing engine implements active revenue management algorithms including time-based decay (e.g., a standard expiry curve), sensory decay modeling, and space-yield displacement, and the algorithms may be applied to all goods stored in the cabinet, including perishable and non-perishable items. In embodiments, the pricing engine implements active revenue management algorithms that go beyond discounts tied solely to printed expiration dates. In one embodiment, the pricing engine implements a sensory decay algorithm that models quality loss occurring before expiration. The system stores, for each SKU or product category, one or more parameters representing predicted loss of sensory appeal (e.g., chocolate blooming, dried fruit clumping, staling, oxidation, moisture-driven texture changes) that may occur prior to the expiration date while the product remains safe to consume. The sensory decay model may use inputs such as time since stocking, lot/batch data, ambient temperature or humidity readings, handling patterns inferred from transactions, and/or visual appearance indicators derived from camera analysis. When the model predicts that continuing to offer the product at full price is likely to reduce sell-through due to diminished sensory appeal, the pricing engine triggers an accelerated discount schedule prior to expiry.

In a further embodiment, the pricing engine implements a **space yield displacement** algorithm that discounts items to free physical slot capacity based on turnover rather than only remaining shelf life. The system computes a revenue-per-slot (or revenue-per-bay) metric over a time window using inventory and transaction logs. If a product remains fresh but occupies a slot with low turnover relative to alternatives, the pricing engine automatically lowers the price to force an exit and free the slot for new, higher-velocity inventory, thereby optimizing physical slot utilization and cabinet throughput. Dynamic pricing decisions can be computed locally by the edge controller so that, even if cloud-based analytics are temporarily unavailable, the system

continues to apply and display appropriate discounts based on locally cached rules and data. Updated prices may be stored in item-level or SKU-level pricing records and communicated to the user interface and, when connectivity exists, to a remote management portal. To guide users to discounted items, the system may activate an addressable indicator (e.g., LEDs) associated with a corresponding shelf zone or module position.

The system can support mixed-mode transactions in which restricted goods and ambient/chilled goods are present in the same cabinet or in adjacent sealed units sharing a common controller, with operating mode, payment logic, biometric requirements, and sensor-fusion thresholds tailored based on the unit type and items enabled for sale. A mode state may be stored in the processing system and optionally controlled by a remote management server. This multi-modal behavior may be implemented in software as a state machine that, in each state, configures device drivers, module sensing logic, pricing rules (including sensory decay and space yield), biometric requirements, payment commitment thresholds, and offline/online behaviors accordingly. Because the claimed system is anchored in specific physical components, as illustrated in FIGS. 1–4—including the secure enclosure with one or more sealed units, the unified reconfigurable chassis supporting interchangeable modules, the induction station for associating SKU and tracking data with internal tags, antenna matrices and/or segmented RFID antennas, camera-based sensor-fusion geometry, load cells for designated zones and weighing dishes, the edge controller with local caching, and access-control hardware—and implements concrete technical logic for sensor fusion, gated gravity sequencing, dynamic pricing based on sensory decay and space yield displacement, offline transaction handling, and restricted-goods smart-glass inspection access, it provides technical improvements over generic retail computers, including improved fraud resistance without robotics, improved inventory rotation, increased sell-through with reduced waste, and resilience to network outages.

VIII. Mobile Ecosystem (Networked Claims). In embodiments, a companion mobile application provides an interactive map of cabinet locations and presents live inventory status for specific cabinets (e.g., quantity remaining for a SKU). The application may present a product “digital twin” for physical units inside the cabinet by displaying batch-specific data including an exact expiration date and a scheduled purge time, and the application mirrors dynamic price changes in substantially real time. In embodiments, the system implements an “ad handshake” protocol that synchronizes context from the mobile application to the cabinet screen, such that a user search or selection in the application (e.g., “cables”) causes the cabinet display to present related promotions and highlight corresponding shelf locations upon the user approaching the cabinet. In embodiments, the application generates last-chance notifications for subscribed items nearing a purge time or reaching a discount threshold.

Claims

1. An autonomous retail system comprising:
 - a secure enclosure having at least one door;
 - an access-control mechanism configured to lock and unlock the door;

a reconfigurable unified chassis within the enclosure defining a plurality of bays configured to simultaneously support a plurality of interchangeable product modules for direct manual retrieval, wherein the chassis includes (i) a vertical power busbar disposed along a rear wall and configured to provide power and data to docked modules at selectable vertical positions, (ii) a mobility system including industrial casters and retractable leveling feet configured to permit undocking for rear-loading FIFO service and to stabilize the cabinet during use, and (iii) a smart edge along a front portion of at least one module comprising a high-density antenna matrix and an LED information strip; the plurality of interchangeable product modules comprising:

(i) a **pusher module** including a spring-loaded track and an adjustable divider configured to advance products toward a user-facing retrieval edge;

(ii) a **hook module** including a hanging rod configured to support suspended products;

(iii) a **scale-box module** defining a product zone supported by at least one load cell; and

(iv) a gated gravity-feed module including an inclined chute configured to direct items to a weighing dish supported by at least one load cell and a controllable gate configured to sequentially release items from the chute into the weighing dish;

(v) a sliding tray module comprising a low-friction gravity shelf configured to advance products toward a retrieval opening without spring compression; and

(vi) an auto-purge module configured to expel an expired or quarantined item from a module position into a smart waste bin.

a sensor network comprising (a) at least one camera oriented to observe an interaction zone adjacent the chassis and (b) at least one item-detection sensor selected from a near-field radio-frequency identification (RFID) antenna, the high-density antenna matrix, or a load cell, wherein the sensor network optionally includes an overhead or interior-facing camera and load cells for restricted-goods zones to enable triple sensor fusion;

a biometric interface including a camera configured to capture facial images of a user;

a user interface configured to receive user input and display transaction information; and

a controller coupled to the sensor network, the biometric interface, the user interface, and the access-control mechanism, the controller being programmed to:

(i) maintain inventory data associating products within the enclosure with product identifiers obtained via the sensor network;

(ii) **detect, using the sensor network, direct manual retrieval of products by a user while the door is unlocked, without robotic retrieval by a gantry**, and determine removed products based on sensor fusion of camera output and the item-detection sensor;

(iii) obtain at least one of expiration data, stocking-time data, environmental data, and sales-velocity or turnover data for products stored in the enclosure, compute dynamic prices based at least in part on (a) a time-based decay curve, (b) a sensory decay model representing predicted quality loss prior to an expiration date, and (c) a space-yield displacement metric indicative of slot stagnation or revenue-per-slot, and control a display or indicator to present discounted pricing for products;

(iv) in a restricted-goods mode, maintain a smart-glass (PDLC) door panel in an opaque sleep state prior to age verification, receive facial image data, determine that the user satisfies a legal age requirement, transition the smart-glass door panel from the opaque sleep state to a transparent inspection state and present product information on the user interface prior to unlocking the door, request a pre-authorization deposit or payment commitment on a payment instrument associated with the user, and unlock the door in response to approval of the pre-authorization deposit or payment commitment, wherein confirmation of restricted-item removal is performed using triple sensor fusion including at least the antenna matrix, an overhead camera, and a load-cell weight delta; and

(v) detect, using the sensor network, removal of at least one product while the door is unlocked, determine a transaction amount for the removed product based on a current dynamic price, and apply a real-time incremental authorization or hold against the payment instrument per removed product to capture the transaction amount from the pre-authorization deposit or payment commitment.

2. The system of claim 1, wherein computing dynamic prices comprises applying a discount function that increases a discount based on the **sensory decay model** indicating reduced sensory quality **before** an expiration date, optionally further based on time remaining until the expiration date, subject to a minimum price threshold.
3. The system of claim 1, wherein controlling the display or indicator comprises illuminating, in a first color or pattern, a zone corresponding to a perishable product whose price has been reduced, and/or displaying a discounted unit price on the user interface.
4. The system of claim 1, wherein the controller is configured to operate in (i) a standard mode in which the access-control mechanism is configured to unlock based on a payment authorization and the biometric interface is bypassed, and (ii) the restricted-goods mode in which the biometric interface, smart-glass inspection state, and pre-authorization deposit or payment commitment are required before unlocking.
5. The system of claim 1, wherein the deposit amount is determined based on a risk profile associated with the user, the risk profile being a function of at least one of prior transaction history, chargeback history, or product category.
6. The system of claim 1, wherein determining that the user satisfies the legal age requirement comprises estimating an age from the facial image data and comparing the estimated age and a confidence measure to configured thresholds.

7. The system of claim 1, wherein determining that the user satisfies the legal age requirement comprises matching the facial image data to a stored user profile having an age attribute that has been verified via a know-your-customer process.
8. The system of claim 1, wherein the controller is configured to synchronize inventory data and pricing parameters with a remote server and to receive updated pricing policies or risk profiles from the remote server.
9. A method of operating an autonomous retail system, the method comprising: detecting, using a biometric interface, a user in proximity to a secure enclosure containing restricted products; capturing facial image data of the user; determining, using an age-verification model executed by an edge processing unit, that the user satisfies a legal age requirement; **maintaining a smart-glass panel of a door in an opaque state prior to the determining; upon determining that the user satisfies the legal age requirement, transitioning the smart-glass panel from the opaque state to a transparent inspection state and presenting product information on a display prior to unlocking the door;** initiating a pre-authorization deposit or payment commitment on a payment instrument associated with the user; upon receiving confirmation of the pre-authorization deposit or payment commitment, unlocking a door of the secure enclosure; detecting, using a sensor fusion region including at least one camera and at least one item-detection sensor, **direct manual retrieval** of at least one product from a module supported by a reconfigurable chassis; determining a transaction amount corresponding to the removed product; and capturing the transaction amount from the pre-authorization deposit or payment commitment.
10. The method of claim 9 further comprising, for perishable non-restricted products stored in the enclosure, obtaining at least one of expiration data, stocking-time data, environmental data, and sales-velocity data, computing dynamic prices based on (i) a **sensory decay model representing predicted quality loss prior to expiration** and (ii) optionally sales-velocity data, and presenting discounted pricing via at least one of an electronic display or an indicator.
11. The system of claim 1, wherein the controller is programmed, for the gated gravity-feed module, to maintain the gate locked while a first item is detected in the weighing dish based on a measured weight exceeding a threshold, to detect removal of the first item based on the measured weight returning below a removal threshold, and **only after** detecting the removal to actuate the gate to release a second item from the chute into the weighing dish.
12. The system of claim 11, wherein the controller confirms the removal of the first item using the camera prior to actuating the gate.
13. The system of claim 1, wherein the controller is programmed to compute a revenue-per-slot metric for at least one module position and to discount a product based on the revenue-per-slot metric to free physical slot capacity for higher-turnover inventory.
14. The system of claim 1, wherein the controller is configured to detect retrieval of an item from the smart waste bin and, in response, capture a transaction amount equal to a full original price of the item to discourage scavenging.
15. The system of claim 1, wherein the controller is configured to monitor load cells for a weight increase indicative of a return event and, subject to a user confirmation of a penalty policy, apply a penalty charge exceeding the unit price of the returned product.

16. The system of claim 1, further comprising a mobile application configured to display batch-specific expiration dates and scheduled purge times for physical units residing in the cabinet, and to synchronize contextual advertising content to the user interface of the cabinet upon detection of the user's mobile device.
17. The system of claim 16, wherein the mobile application transmits user search history or preferences to the controller, and the controller updates the cabinet display to highlight products matching said history upon user approach.
18. The system of claim 1, wherein the gated gravity-feed module is configured to lock the gate in a closed position while the load cell detects weight in the weighing dish, preventing release of a subsequent item until the weighing dish is emptied.
19. The system of claim 1, wherein the vertical power busbar comprises conductive rails extending along a rear wall and wherein each interchangeable product module comprises a mating connector configured to receive power and exchange data when docked to the rails at a selected vertical position.
20. The system of claim 1, wherein the mobility system comprises industrial casters and retractable leveling feet, and wherein the controller is configured to present a service mode enabling rear-loading FIFO stocking when the cabinet is undocked.
21. The system of claim 1, wherein the smart edge comprises the high-density antenna matrix configured for virtual zoning and an LED information strip configured to display at least one of per-item guidance, discounted pricing cues, or navigation prompts.
22. The system of claim 1, wherein the sliding tray module comprises a gravity shelf having a low-friction floor and is configured to store soft packaging items that would be damaged by spring compression.
23. The system of claim 1, wherein the auto-purge module comprises a retractable front stop configured to release an item upon a purge condition and to direct the item into the smart waste bin.
24. The system of claim 1, wherein the smart waste bin comprises at least one load cell and an RFID antenna, and wherein items placed in the smart waste bin are tracked as quarantined pending staff collection.
25. The system of claim 6, wherein if a user retrieves an item from the smart waste bin, the controller applies an anti-scavenger charge corresponding to an original price associated with the retrieved item.
26. The system of claim 1, wherein at least one of the pusher module or the hook module comprises a one-way ratchet configured to prevent rearward movement of a product toward a rear-loading side after the product is advanced toward the retrieval edge.
27. The system of claim 1, wherein the controller is programmed, for a weight-based zone, to detect a return event based on a load-cell weight increase during an active session and to apply a penalty charge that is greater than a current unit price, and wherein the user interface requires a user consent input acknowledging the penalty policy prior to unlocking the door.
28. The system of claim 1, wherein prior to the payment commitment or door unlock, the user interface presents an exact expiration date and a scheduled purge time for at least one physical unit inside the enclosure.
29. The system of claim 1, wherein the controller applies incremental authorizations or holds per removed item in substantially real time as removals are confirmed by sensor fusion.

30. The system of claim 1, further comprising a companion mobile application configured to present live inventory status for a cabinet, to mirror dynamic price changes in substantially real time, to present batch-specific digital twin information including an expiration date and a purge timer, and to provide last-chance notifications for items nearing purge.

Abstract

A modular smart vending cabinet system includes a secure enclosure with a lockable door, an edge controller, and a reconfigurable unified chassis supporting multiple interchangeable product modules operable simultaneously for direct manual retrieval. The chassis includes a vertical power busbar along a rear wall for tool-less vertical repositioning of modules, and a mobility system with industrial casters and retractable leveling feet to enable undocking for rear-loading FIFO service. Module types include spring-loaded pushers with adjustable dividers, hooks or rods for hanging items, box-on-scale zones using load-cell weight deltas corroborated by camera evidence, sliding trays with low-friction floors for soft packaging, and gated gravity-feed channels in which a controllable gate releases a next item into a weighing dish only after confirmed removal of a prior item. In a restricted-goods mode, a PDLC smart-glass panel defaults to an opaque state for privacy and transitions to a transparent inspection state after biometric age verification to present product information before access, and restricted-item removal is confirmed using triple sensor fusion including an antenna matrix, an overhead camera, and load cells. A pricing engine applies dynamic discounts using time-based decay, sensory decay modeling of quality loss prior to expiration, and space-yield displacement based on slot turnover, and transactions may be captured via real-time incremental holds per item removed. The system may enforce a sales-final policy via mechanical anti-reverse constraints and software penalties, automatically purge expired items into a smart waste bin, and provide a mobile application with live inventory, batch-specific digital twin data, price mirroring, and purge/discount notifications.